

4 Hypothesis tests

What students need to learn:

Population, census and sample. Sampling unit, sampling frame.

Students will be expected to know the advantages and disadvantages associated with a census and a sample survey.

Concepts of a statistic and its sampling distribution.

Concept and interpretation of a hypothesis test. Null and alternative hypotheses.

Use of hypothesis tests for refinement of mathematical models.

Critical region.

Use of a statistic as a test statistic.

One-tailed and two-tailed tests.

Hypothesis tests for the parameter p of a binomial distribution and for the mean of a Poisson distribution.

Students are expected to know how to use tables to carry out these tests. Questions may also be set not involving tabular values. Tests on sample proportion involving the normal approximation will not be set.

Unit S3

Statistics 3

GCE AS and GCE Further Mathematics and GCE AS and GCE Further Mathematics (Additional) A2 optional unit

S3.1 Unit description

Combinations of random variables; sampling; estimation, confidence intervals and tests; goodness of fit and contingency tables; regression and correlation

S3.2 Assessment information

Prerequisites

A knowledge of the specifications for S1 and S2 and their prerequisites and associated formulae is assumed and may be tested.

Examination

The examination will consist of one 1½ hour paper. The paper will contain about seven questions with varying mark allocations per question which will be stated on the paper. All questions may be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: +, −, ×, ÷, π, x^2 , $\sqrt{x}$, $\frac{1}{x}$, x^y , $\ln x$, e^x , $x!$, sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Students are expected to know any other formulae which might be required by the specification and which are not included in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Students will be expected to know and be able to recall and use the following formulae:

$aX \pm bY \sim N(a\mu_x \pm b\mu_y, a^2\sigma_x^2 + b^2\sigma_y^2)$ where X and Y are independent and $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$.

1 Combinations of random variables

What students need to learn:

Distribution of linear combinations of independent Normal random variables.

If $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$

independently, then

$$aX \pm bY \sim N(a\mu_x \pm b\mu_y, a^2\sigma_x^2 + b^2\sigma_y^2).$$

No proofs required.

2 Sampling

What students need to learn:

Methods for collecting data. Simple random sampling. Use of random numbers for sampling.

Other methods of sampling: stratified, systematic, quota.

The circumstances in which they might be used. Their advantages and disadvantages.

3 Estimation, confidence intervals and tests

What students need to learn:

Concepts of standard error, estimator, bias.

The sample mean, $\bar{x}$, and the sample variance,

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \text{ as}$$

unbiased estimates of the corresponding population parameters.

The distribution of the sample mean $\bar{X}$.

$\bar{X}$ has mean μ and variance $\frac{\sigma^2}{n}$.

If $X \sim N(\mu, \sigma^2)$ then $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$.

Concept of a confidence interval and its interpretation.

Confidence limits for a Normal mean, with variance known.

Hypothesis tests for the mean of a Normal distribution with variance known.

Use of Central Limit theorem to extend hypothesis tests and confidence intervals to samples from non-Normal distributions. Use of large sample results to extend to the case in which the variance is unknown.

Hypothesis test for the difference between the means of two Normal distributions with variances known.

Use of large sample results to extend to the case in which the population variances are unknown.

Link with hypothesis tests.

Students will be expected to know how to apply the Normal distribution and use the standard error and obtain confidence intervals for the mean, rather than be concerned with any theoretical derivations.

Use of $\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim N(0, 1)$.

$\frac{\bar{X} - \mu}{S/\sqrt{n}}$ can be treated as $N(0, 1)$ when n is large. A knowledge of the t -distribution is not required.

Use of
$$\frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_y)}{\sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}} \sim N(0, 1).$$

Use of
$$\frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_y)}{\sqrt{\frac{S_x^2}{n_x} + \frac{S_y^2}{n_y}}} \sim N(0, 1).$$

A knowledge of the t -distribution is not required.